

Hands-On-Learning: : Automate Governance Monitoring

Learning Objectives

By the end of this activity, learners will be able to:

- Conceptualize an AI-driven workflow to automate a routine governance monitoring task.
 - Use a generative AI tool (Google Gemini) to analyze security logs and generate a clear, human-readable summary.
 - Translate unstructured data (log entries) into actionable governance artifacts (risk summaries and project tasks).
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Description

In this final lab, learners act as a cloud security engineer looking to innovate their way out of manual, repetitive work. The activity focuses on using AI to automate the time-consuming process of reviewing security logs, identifying risks, and creating remediation tasks. This exercise demonstrates the principles of continuous improvement and optimization, showing how modern tools can be used to make a governance program more efficient, responsive, and intelligent.

Structure of the Activity

Title: Automate Governance Monitoring with AI

Tool used:

Google Gemini

Scenario

You are a Cloud Security Engineer at "PixelStream," a digital media company. You are overwhelmed by the volume of daily security alerts from your cloud infrastructure. Manually reviewing logs for suspicious activity and then creating tickets is taking up half your day. You decide to build a proof-of-concept AI workflow using Google Gemini to automate this process, turning raw log data into structured, actionable intelligence.

Objective

To explore AI applications by designing a workflow in Google Gemini that continuously monitors infrastructure (simulated), gathers evidence from logs, and auto-generates a clear risk summary and actionable project-management tasks.



Instructions for learners

1 Simulate the Input:

Copy the following sample log entries. This will be your input data for Gemini.

[2023-10-27 10:05:14] INFO: User 'admin@pixelstream.com' logged in from IP 8.8.8.8.

[2023-10-27 10:15:22] WARN: Multiple failed login attempts for user 'root' from IP 192.168.1.10.

[2023-10-27 10:17:45] CRITICAL: Port scan detected from IP 123.45.67.89 on server 'db-prod-01'.

[2023-10-27 10:21:03] INFO: User 'jane@pixelstream.com' accessed file 'Q3-financials.pdf'.

2 Design the AI Workflow:

Open Google Gemini. You will use a multi-step prompt to create your workflow.

Step 1 - Analysis Prompt: Paste the log data into Gemini and use this prompt: "Act as a senior security analyst. Analyze the following log entries. Identify any security events that represent a potential risk."

Step 2 - Reporting Prompt: Follow up with this prompt: "Based on your analysis, generate a brief, prioritized risk summary report. For each event, list its risk level (Low, Medium, High, Critical), a short description, and a recommended action."

Step 3 - Action Prompt: Finally, use this prompt: "For the 'Critical' risk event, create a project management task with a title and a description for a junior analyst to investigate."

3 Review the Output:

Examine the risk summary report and the project task generated by Gemini.

Note how the AI successfully transformed unstructured text into a prioritized summary and then into an actionable instruction.

Organize Information Following This Structure for Peer Review

Title Page: Project title, your name, date.

AI Workflow and Output: Submit a document containing:

- 1 The three prompts you used in sequence.
- 2 The final risk summary report generated by Gemini.

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The final project management task generated by Gemini.

Automation Benefit: In a short paragraph, explain how this AI-powered workflow would improve the efficiency and responsiveness of the security team at PixelStream.

Instructions for Documenting and Sharing the Project for Peer Review

Document the Project



Use word processing software to create your report. Copy and paste the prompts and AI outputs directly into your document. Ensure all sections are completed.

Share the Project



Save your report in PDF format. Upload your document to the “Peer Review” area in the course shell. Provide a brief description of your project in the submission post.



Peer Review

Review the projects submitted by your peers and provide constructive feedback on their work.

This activity gives learners a chance to apply the lesson by designing an automated workflow that embodies the principle of continuous improvement, using AI to create a more scalable and intelligent governance monitoring system.